TSM 099 Arithmetic Test

Practice 1 | Version 202610122020.16

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Evaluate the principal square root in each part. $(a)\quad \sqrt{25}$ $(b)\quad \sqrt{-49}$	1. (a) 10 pts (b) 10 pts
2. Evaluate each side separately at x = 6, -6, 0. Record the left value, right value, and whether they are equal. State what the tested values establish; do not solve the equation. $\frac{x^2-4}{x-2} = \frac{x^2+5x+6}{x+3}$	2. x = 6: left / right / equal? 4 pts x = -6: left / right / equal? 4 pts x = 0: left / right / equal? 4 pts Conclusion 8 pts
3. Evaluate at (a) x = 2 and (b) x = -2. Show substitution. $x^2 - 3x + 2$	3. (a) 10 pts (b) 10 pts
4. Reduce each fraction to lowest terms. $(a)\quad \frac{14}{21}$ $(b)\quad \frac{42}{66}$	4. (a) 10 pts (b) 10 pts
5. Simplify the square root. $\sqrt{18}$	5. Radical 20 pts

Instructor answer key: TSM 099 Arithmetic Test

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a)

5

10 pts

(b)

7i

10 pts

2. x = 6: left / right / equal?

Left 8; right 8; yes

4 pts

x = -6: left / right / equal?

Left -4; right -4; yes

4 pts

x = 0: left / right / equal?

Left 2; right 2; yes

4 pts

Conclusion

All tested values work; this alone does not prove an identity.

8 pts

3. (a)

0

10 pts

(b)

12

10 pts

4. (a)

$\frac{2}{3}$

10 pts

(b)

$\frac{7}{11}$

10 pts

5. Radical

$3\sqrt{2}$

20 pts